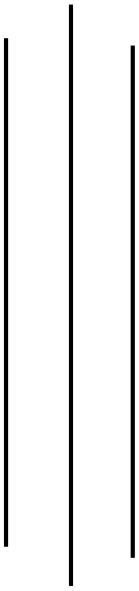
Lumbini Technological University
Institute of Engineering and Information Technology

Banke, Lumbini Province

Syllabus



B.Tech in CS & AI

Year/Semester: II/II

Research Methodology

Course Code	BAI251	**Year/Semester**	II/II
Credit Weightage	3	**Lecture**	3 hrs/wk
		Tutorial	1 hr/wk
		Practical	-
		Total	64 hrs

1. Course Objective

This course aims to provide students with essential research skills, covering research design, data collection, and analysis. It integrates both qualitative and quantitative methods, emphasizing practical application through case studies and real-world examples. Students will learn to craft research proposals, analyze data, and present findings, preparing them for academic and professional research in civil engineering.

2. Course Detail

2.1 Theory (48 hrs)

Unit 1: Basics of research (4 hrs)

Introduction to research: Definition and importance. Types of research: Qualitative, quantitative, mixed methods. Research process: Steps in conducting research. Ethical considerations in research: Informed consent and avoiding plagiarism. Intellectual property rights: Patents, copyrights, trademarks, and protecting research work.

Unit 2: Research problem and literature review (7 hrs)

Identifying research problems: Characteristics of a good research problem, formulating research questions. Literature review: Purpose and importance, conducting a literature search, writing a literature review.

Unit 3: Research design and method (12 hrs)

Research design basics: Types of research designs (experimental, descriptive, correlational, exploratory), choosing an appropriate research design. Case study and survey research: Designing case studies, designing surveys and questionnaires. Qualitative methods: Interviews and focus groups, observation techniques. Qualitative data analysis: Coding and thematic analysis, software tools for qualitative analysis. Quantitative methods: Surveys and questionnaires, experimental data collection. Quantitative data analysis: Descriptive and inferential statistics, using software tools (e.g., SPSS, Excel). Mixed methods research: Introduction to mixed methods, integration of qualitative and quantitative data, designing mixed methods research, data collection and analysis in mixed methods (strategies for data integration).

Unit 4: Advanced topics in research methodology (9 hrs)

Introduction to meta-analysis: Purpose and importance, conducting a meta-analysis. Systematic reviews: Methodology and best practices, examples and applications. Advanced sampling methods: Probability and non-probability sampling, sample size determination. Advanced sampling designs:

Stratified, cluster, and systematic sampling. Introduction to big data: Characteristics and challenges. Data mining techniques: Association rules, classification, and clustering.

Unit 5: Research proposal and report writing (9 hrs)

Components of a research proposal: Title, abstract, and introduction; literature review and methodology. Writing a research proposal: Tips for effective proposal writing, examples and templates. Structure of a research report: Introduction, methodology, results, discussion, conclusions, and recommendations. Writing and presenting research findings: Effective writing strategies, preparing and delivering presentations.

Unit 6: Writing and publishing research papers (7 hrs)

Structure of a research paper: Abstract, introduction, methodology, results, discussion, and conclusion. Writing tips and strategies: Clarity, coherence, conciseness, and common pitfalls and how to avoid them. Understanding the peer review process: Types of peer review, responding to reviewers' comments. Selecting a journal for publication: Factors to consider, submission guidelines and process.

2.2 Tutorials

Unit-specific exercises focused on case studies and applications in Civil Engineering, leading to a research paper draft.

2.3 Laboratory Work

N/A

3. Evaluation Scheme

Evaluation of students' performance is divided into two main parts: (1) internal evaluation and (2) semester examination, with breakdown as follows.

Evaluation	Components	Weightage	Full Marks	Pass Marks
Internal evaluation	Class attendance and performance	5	40	20
	Assignment	15		
	Seminar/project/presentation	20		
	Practical/lab examination	-	-	-
	Internal examination	**20**	**20**	**10**
	Total of internal evaluation		**60**	**30**
Semester examination			**40**	**20**

5. Books

- C.R. Kothari. "Research Methodology - Methods and Techniques." New Age International.
- D. V. Thiel, "Research Methods for Engineers." Cambridge University Press.
- J. W. Creswell and J. D. Creswell, "Research Design: Qualitative, Quantitative, and Mixed Methods Approaches." SAGE Publications.
- R. Kumar, "Research Methodology: A Step-by-Step Guide for Beginners." SAGE Publications.

Python Programming

Course Code	BAI252	**Year/Semester**	II/II
Credit Weightage	3	**Lecture**	3 hrs/wk
		Tutorial	1 hrs/wk
		Practical	2 hrs/wk
		Total	96 hrs

1. Course Description

This Programming with Python course is designed to provide students with a comprehensive understanding of Python programming and web development using the framework. The course starts with an introduction to Python, covering essential topics such as variables, data types, control flow, and functions. Students will learn how to handle files and gain a solid understanding of object-oriented programming (OOP) in Python.The course then delves into a popular web development framework in Python. Students will learn how to set up a python project, create models, perform database operations, and handle user authentication and authorization. Additionally, students will explore templating engine, forms, and the admin interface. Optional topics include Rest Framework for building RESTful APIs and deploying the application on web servers like Heroku or AWS.Throughout the course, students will work on a mini-project using Framework, applying the concepts they have learned to develop a small-scale web application with dynamic content rendering, user authentication, and database interactions.

2. Course Objectives

The primary objective of this course is to equip students with the knowledge and skills to become proficient Python programmers and web developers using the framework. By the end of the course, students should be able to:

- Understand the fundamentals of Python programming, including variables, data types, control flow, and functions.
- Grasp the concepts of object-oriented programming in Python and apply them to real-world scenarios.
- Develop web applications including setting up a project, creating models, and handling user authentication and authorization.
- Utilize Framework templating engine and forms to render dynamic content and handle user input.

3. Course Details

3.1 Theory (48 hrs)

Unit 1: Python Basics (14 hrs)

Introduction to Python: Overview of Python and its applications, Installing Python and setting up the environment Running, Python scripts and interactive mode. Variables and Data Types: Numeric data types (int, float, complex), String data type and manipulation, tuples, and dictionaries, Boolean values and operators. Control Flow: If-else statements Loops (for and while), Break and continue statements, Exception handling with try-except blocks. Functions: Defining functions and function arguments, Return values and function scopes, Lambda functions and higher-order functions, Recursion. File Handling: Reading and writing text files, Working with CSV and JSON files

Unit 2: Python OOP **(9 hrs)**

Introduction to Object-Oriented Programming (OOP): Understanding classes and objects, Encapsulation, inheritance, and polymorphism, Constructors and destructors. Working with Classes and Objects: Defining classes and creating objects Instance and class variables, Methods and properties Inheritance and Composition Extending classes through inheritance
Composition and using multiple classes together. Advanced OOP Concepts: Method overloading and overriding, Abstract classes and interfaces, Class decorators

Unit 3: Python Scripting **(5 hrs)**

Command-Line Arguments, Parsing command-line arguments, Arg parse module for more complex argument parsing. Working with Modules: Creating and importing modules Packages and the module search path. Regular Expressions: Pattern matching using regular expressions, Replacing and searching for patterns

Unit 4: Python Database and Database Connectivity **(6 hrs)**

Introduction to Databases: Understanding relational databases and SQL Overview of popular database systems. SQL Basics: CRUD operations (Create, Read, Update, Delete) Joins, subqueries, and aggregate functions Database Connectivity in Python: Connecting to databases from Python, ExecutingSQL queries with Python

Unit 5: Introduction to Web Development **(14 hrs)**

Understanding web development concepts, Introduction to Framework and its features, setting up any Framework project and running the development server. Framework Models: Defining models and database tables, Querying data with Framework ORM (Object-Relational Mapping). Views and Templates: Creating views and handling URL patterns Using any Framework templating engine Rendering dynamic data in templates Forms and User Input: Handling forms and user input in Framework Form validation and processing Authentication and Authorization: Managing user accounts and authentication Controlling access to views and resources Static Files and Media Handling: Managing static files (CSS, JavaScript, etc.). Uploading and serving media files (images, videos, etc.). Admin Interface: Using any Framework built-in admin interface Customizing the admin interface for models. Rest Framework (optional): Introduction to RESTful APIs, Creating APIs with Rest Framework

3.2 Tutorials

Unit-specific exercises focused on problem-solving.

3.3 Laboratory Works

- Project: Personal Blogging Platform
- Description: In this project, you will build a personal blogging platform using Framework. The platform will allow users to register, log in, create blog posts, and view posts from other users. Users can like and comment on blog posts, and admins can manage the posts and user accounts through the admin interface.

4. Evaluation Scheme

Evaluation of students' performance is divided into two main parts: (1) Internal evaluation and (2) Semester examination, with breakdown as follows.

Evaluation	Components	Weightage	Full Marks	Pass Marks
Internal Evaluation	Class Attendance and Performance	5	20	10
	Assignment	10		
	Seminar/Project/Presentation	5		
	Practical/Lab Examination	20	20	10
	Internal Examination	**20**	**20**	**10**
	Total Internal Marks		**60**	**30**
Semester Examination			**40**	**20**

5. Books

- "Python Crash Course" by Eric Matthes (2nd Edition): ISBN-13: 978-1593279288
- "Django for Beginners" by William S. Vincent (3rd Edition): ISBN-13: 978-1099701185
- "Python Cookbook" by David Beazley and Brian K. Jones
- "Django 3 By Example" by Antonio Mele

Theory of computation

Course Code	BAI53	**Year/Semester**	II/II
Credit Weightage	3	**Lecture**	3 hrs/wk
		Tutorial	1 hrs/wk
		Practical	2 hrs/wk
		Total	96 hrs

1. Course Description

This course is designed to provide students with a comprehensive understanding of concepts of theory of computation. The course covers different aspects including string operations, language constructs, finite automata, regular expression, context free language, push down automata, and Turing machine. Through a combination of lectures, practical exercises, and projects, students will gain the necessary skills to design and develop automata machines.

2. Course Objectives

The course aims to achieve the following objectives:

- Introduce different concepts of automata theory
- Provide concepts and skills of designing finite machines
- Familiarize with regular expressions
- Provide concepts of designing grammar
- Familiarize with formal language classes
- Present concepts of push down automata
- Acquaint with working mechanism of Turing Machines

3. Course Details

3.1 Theory (48 hrs)

Unit 1: Introduction (4 Hrs.)

Complexity Theory, Computability Theory, Automata Theory, Determinism and Non-determinism, Alphabets, Kleen and Positive of Closure Alphabets, Strings, Empty String, Suffix, Prefix and Substring of a string, Concatenation of strings, Languages, Empty Language

Unit 2: Finite Automata (9 Hrs.)

Finite Automata, Finite State Machine, Deterministic Finite Automata (DFA), Notations for DFA, Language of DFA, Extended Transition Function of DFA, Non-Deterministic Finite Automaton (NFA), Notations for NFA, Language of NFA, Extended Transition Function of NFA, Equivalence of DFA and NFA, Subset-Construction Method, Method for reduction of NFA to DFA, Theorems for equivalence of Language accepted by DFA and NFA, Finite Automaton with Epsilon Transition (ε - NFA), Notations for ε - NFA, Epsilon Closure of a State, Extended Transition Function of ε – NFA, Equivalence of NFA and ε –NFA, Equivalence of DFA and ε – NFA, Finite State Machines with output: Moore Machine and Mealy Machines

Unit 3: Regular Expressions and Regular Languages (8 Hrs.)

Regular Expressions, Operators in Regular Expressions (Union, Concatenation, Kleen), Algebraic Rules for Regular Expressions, Application of Regular Expressions, Equivalence of Regular Expression and Finite Automata, Reduction of Regular Expression to ε–NFA, Conversion of DFA to Regular Expression, Arden's Theorem, Regular Languages, Properties of Regular Languages, Pumping Lemma for Regular Expression, Closure Properties of Regular Languages, Minimization of Finite State Machines

Unit 4: Context Free Grammars and Context Free Languages (11 Hrs.)

Context Free Grammar (CFG), Components of CFG, Use of CFG, Context Free Language (CFL), Bottomup and Topdown derivation approaches, Leftmost and Rightmost derivations, Sentential Form (Left, Right), Language of a grammar, Parse tree and its construction, Ambiguous grammar, Inherent Ambiguity, Right Linear and Left Linear regular grammars, Equivalence of regular grammar and finite automata, CFG Simplification: Removal of Useless symbols, Nullable Symbols, and Unit Productions, Chomsky Normal Form (CNF), Greibach Normal Form (GNF), Backus-Naur Form (BNF), Context Sensitive Grammar, Chomsky Hierarchy(Type 0, 1, 2, 3), Pumping Lemma for CFL, Closure Properties of CFL

Unit 5: Push Down Automata (7 Hrs.)

Push Down Automata (PDA), Representation of PDA, Operations of PDA, Moves of a PDA, Instantaneous Description for PDA, Deterministic PDA, Non Deterministic PDA, Acceptance of strings by PDA, Language of PDA, PDA by Final State, PDA by Empty Stack, Conversion of PDA by Final State to PDA accepting by Empty Stack and vice-versa, Conversion of CFG to PDA and vice-versa

Unit 6: Turing Machine (9 Hrs.)

Introduction to Turing Machines (TM), Notations of Turing Machine, Language of a Turing Machine, Instantaneous Description for Turing Machine, Acceptance of a string by a Turing Machines**,** Turing Machine as a Language Recognizer, Turing Machine as a Computing Function, Turing Machine with Storage in its State, Turing Machine as a enumerator of stings of a language, Turing Machine as Subroutine**,** Turing Machine with Multiple Tracks, Turing Machine with Multiple Tapes, Equivalence of Multi-tape Turing Machine and Multi-track Turing Machine, Non-Deterministic Turing Machines, Restricted Turing Machines: With Semi-infinite Tape, Multistack Machines, and Counter Machines**,** Curch Turing Thesis, Universal Turing Machine, Turing Machine and Computers, Encoding of Turing Machine, Enumerating Binary Strings, Codes of Turing Machine, Universal Turing Machine for encoding of Turing Machine

3.2 Tutorials

Unit specific problem solving excises.

3.2 Laboratory Work

The practical work includes implementing the concepts like

- string computation,
- DFA and NFA,

- Tokenizers for identifiers, keywords, etc.
- Regular expressions,
- Grammar derivations,
- Parse tree construction,
- PDA with finals state,
- PDA with empty stack
- Turing Machine.

4. Evaluation Scheme

Evaluation of students' performance is divided into two main parts: (1) internal evaluation and (2) semester examination, with breakdown as follows.

Evaluation	Components	Weightage	Full Marks	Pass Marks
Internal Evaluation	Class Attendance and Performance	5	20	10
	Assignment	10		
	Seminar/Project/Presentation	5		
	Practical/Lab Examination	20	20	10
	Internal Examination	**20**	**20**	**10**
	Total Internal Marks		**60**	**30**
Semester Examination			**40**	**20**

5. Books

- Introduction to Automata Theory, Languages, and Computation; John E. Hopcroft, Rajeev Motwani, Jeffrey D. Ullman; 3rd Edition; Pearson - Addison-Wesley.
- Automata, Computability and Complexity: Theory and Applications; Elaine Rich; Pearson
- Introduction to the Theory of Computation; Michael Sipser; 3rd Edition; Cengage Learning
- What Can Be Computed?: A Practical Guide to the Theory of Computation; John MacCormick; Princeton University Press
- Theory of Computing: A Gentle introduction; Efim Kinber, Carl Smith; Prentice- Hall.
- Introduction to Languages and the Theory of Computation; John Martin; Tata McGraw Hill.
- *Elements of the Theory of Computation;* Harry R. Lewis and Christos H. Papadimitriou; Prentice Hall.

Numerical Method

Course Code	BAI254	**Year/Semester**	II/II
Credit Weightage	3	**Lecture**	3 hrs/wk
		Tutorial	1 hrs/wk
		Practical	2 hrs/wk
		Total	96 hrs

1. Course Description

Numerical methods are those techniques that give an approximate solution of a mathematical problem that has no easy way to calculate an analytical solution. This course consists of the study of numerous tools and techniques required to find numerical solutions of various mathematical problems that frequently arise in the field of science and engineering. Together with the theory, implementation of the theory by means of practical classes will further strengthen the grasp of the subject material.

2. Course Objectives

The objective of this course is to

- Make clear why numerical techniques are needed and the conditions under which such techniques have to be implemented.
- Give an introductory idea about the errors that can arise during numerical computation.
- Introduce various numerical methods for solving a nonlinear equation and comparison of those methods in terms of efficiency.
- Provide the necessary methods of interpolation and approximation and the ability to differentiate between the two.
- Give necessary tools and techniques to solve linear algebraic equations using direct as well as iterative methods.
- Develop the skills to use various techniques of numerical differentiation and integration.
- Introduce the numerical methods of solving first and higher order ordinary differential equations that are specified using initial conditions and boundary conditions.
- Give some methods of numerical solutions of Laplace's equation and Poisson's equation.

3. Course Details

3.1 Theory (48 hrs.)

Unit 1: Numerical Solution of Nonlinear Equations (9 hrs)

Taxonomy Errors, computing error, Intermediate Value Theorem, Rolle's Theorem, Mean Value Theorem, Taylor's Series

Solving non-linear equations: Bisection Method, Secant Method, Newton Raphson Method, Fixed Point Method and False Position method and their convergence.

Unit 2: Interpolation and Least Squares Approximation (8 hrs)

Linear Interpolation, Lagrange's Interpolation Polynomial, Divided, Newton's Interpolation and its Derivation, Divided Difference Table, Error in Polynomial Interpolation, Interpolation for Evenly Spaced Data: Newton's Forward and Backward Difference Interpolation and their Derivation, Cubic Spline Interpolation using Natural Cubic Splines

Least Squares Approximation, Method of Linear Least Squares and its Derivation, Method of Nonlinear Least Squares with Derivation (Exponential and Polynomial Curve Fitting)

Unit 3: Solution of Linear Algebraic Equations (10 hrs)

Review of System of Linear Equations, Consistent and Inconsistent Systems, Gaussian Elimination Algorithm, Ill Conditioned System and Partial Pivoting, Gauss-Jordan Algorithm, Matrix Inversion Method, Matrix Factorization Methods (Dolittle, Crout and Cholesky), Diagonally Dominant System, Iterative Methods (Jacobi, Gauss-Seidel), Convergence of Iterative Methods for Diagonally Dominant Systems, Eigenvalues and Eigenvectors, Eigenvalue Problem, Solving Eigenvalue Problem using Power Method

Unit 4: Numerical Differentiation and Integration (7 hrs)

Numerical Differentiation, Forward Difference, Backward Difference and Central Difference Formula for First Derivative, Interpolation Formula for Derivatives, Error Analysis, Central Difference Formula for Second Derivative

Numerical Integration, Trapezoidal Rule with Derivation and Error, Simpson's 1/3 Rule with Derivation and Error, Simpson's 3/8 Rule with Derivation and Error, Composite Rules for Trapezoidal, Simpson's 1/3 and Simpson's 3/8 Methods and their Derivation, Romberg Integration, Gaussian Quadrature Formula with Derivation, Double Integrals

Unit 5: Ordinary Differential Equations (10 hrs)

Review of ODE, Order and Degree, Initial and Boundary Value Problems, First-Order First-Degree Equations and their Solutions (Taylor's Series Method, Euler's Method, Modified Euler's Method, Runge-Kutta Method), Solving upto Higher Order Equations, Solving Boundary Value Problem using Shooting Method

Unit 6: Partial Differential Equations (4 hrs)

Review of PDE, Elliptic, Parabolic and Hyperbolic PDE, Laplace's Equation and its Difference Equation, Poisson's Equation and its Difference Equation, Solution of Laplace's and Poisson's Equation Using Finite Difference Method

3.2 Tutorial

Unit-specific exercises focused on problem-solving

3.3 Laboratory Works

The practical work will consist of the implementation of methods and algorithms using any programming language. The practical work should include the input-output analysis and demonstrate the understanding of change in accuracy, number of iterations etc. for changes in different parameters in the algorithm.

The practical work should include the implementation of

- Bisection method, Secant method, Newton Raphson method, Fixed-point iteration method and false position method.
- Lagrange's interpolation, Newton's interpolation using divided differences, Interpolating evenly-spaced data using forward difference and backward difference.

- Linear least squares approximation, Non-linear least squares approximation (Polynomial and Exponential fitting).
- Gaussian elimination algorithm, Gaussian elimination algorithm with partial pivoting, Gauss-Jordan algorithm, Jacobi method, Gauss-Seidel method, Power method.
- Central difference method for first and second derivatives, Interpolation method for first derivative.
- Composite trapezoidal method, Composite Simpson's 1/3 method and Composite Simpson's 3/8 method.
- Euler's, Modified Euler's and Runge-Kutta method for solving first order differential equations, Solving systems of differential equations and second order equations using modified Euler's method, Shooting method.
- Implementation of Laplace and Poisson's equation.

4. Evaluation Scheme

Evaluation of students' performance is divided into two main parts: (1) Internal evaluation and (2) Semester examination, with breakdown as follows.

Evaluation	Components	Weightage	Full Marks	Pass Marks
Internal Evaluation	Class Attendance and Performance	5	20	10
	Assignment	10		
	Seminar/Project/Presentation	5		
	Practical/Lab Examination	20	20	10
	Internal Examination	**20**	**20**	**10**
	Total Internal Marks		**60**	**30**
Semester Examination			**40**	**20**

5. Books

- Balagurusamy, E. Numerical Methods, McGraw Hill Education.
- B. S. Grewal, Numerical Methods in Engineering and Science, Khanna Publications.
- Cheney, W. and Kincaid, D. Numerical Mathematics and Computing, 7th Edition, Cengage Learning.
- Burden, R. L. and Faires, J. D. Numerical Analysis, 9th Edition, Cengage Learning.
- Chapra, S. C. and Canale, R. P. Numerical Methods for Engineers, 7th Edition, McGraw Hill Education.
- Gerald, C. F. and Wheatley, P. O. Applied Numerical Analysis, 6th Edition, Pearson Education.

Software Engineering

Course Code	BAI255	**Year/Semester**	II/II
Credit Weightage	3	**Lecture**	3 hrs/wk
		Tutorial	1 hrs/wk
		Practical	2 hrs/wk
		Total	96 hrs

1. Course Description

This course is designed to provide students with a fundamental understanding of state of arts of Software Engineering. The course covers different aspects of software development, software maintenance and software trends in software industries.

2. Course Objectives

The course aims to achieve the following objectives:

- Introduce students to the fundamental concepts and technologies that underpin the Software Engineering
- Provide a solid basic foundation of Software Development approaches.
- Familiarize students with different software development approaches.

3. Course Detail

3.1 Theory (48 Hrs)

Unit 1. Introduction (3 Hrs.)
Professional Software Development; Software Engineering Ethics

Unit 2: Software Processes (5 Hrs.)
Software Process Models; Process Activities; Coping with Change; Process Improvement

Unit 3: Agile Software Development (4 Hrs.)
Agile Methods; Agile Development Techniques

Unit 4: Requirements Engineering (4 Hrs.)
Functional and Non-functional Requirements; Requirements Engineering Processes; Requirements Elicitation; Requirements Specification; Requirements Validation; Requirements Change

Unit 5: System Modeling (6 Hrs.)
Context Models; Interaction Models; Structural Models; Behavioral Models; Model-Driven Architecture

Unit 6: Architectural Design (5 hrs.)
Architectural Design Decisions; Architectural Views; Architectural Patterns; Application Architectures

Unit 7: Design and Implementation (6 Hrs.)
Object-oriented Design using the UML; Design Patterns; Implementation Issues; Open-source development

Unit 8: Software Testing **(4 Hrs.)**
Development Testing; Test-driven Development; Release Testing; User Testing

Unit 9: Software Evolution **(4 Hrs.)**
Evolution Processes; Legacy Systems; Software Maintenance

Unit 10: Software Management **(7 Hrs.)**
Project management (Risk Management, Managing People, Teamwork); Project Planning (Software Pricing, Plan-driven Development, Project Scheduling, Estimation Techniques, COCOMO); Quality Management (Software Quality, Software Standards, Reviews and Inspections, Quality Management and Agile Development, Software Measurement); Configuration management (Version Management, System Building, Change Management, Release Management).

3.2 List of Tutorial

Unit-specific exercises focused on problem-solving

3.3 Laboratory Work

The practical work includes following activities:

1. Writing the problem statement for any system.
2. Preparing software requirement specification (SRS) for any system.
3. Drawing all UML diagrams using appropriate tools.
4. Learning to use CASE and project management tools.
5. Learning to design and run the test cases using appropriate tools.

In addition to above mentioned activities, students should prepare project report for some system including different activities of software process model.

4. Evaluation Scheme

Evaluation of student's performances is divided into two main parts: (1) Internal Evaluation and (2) Semester examination, with breakdowns as follows:

Evaluation	**Components**	**Weightage**	**Full Marks**	**Pass Marks**
Internal Evaluation	Class Attendance and Performance	5	20	10
	Assignment	10		
	Seminar/Project/Presentation	5		
	Practical/Lab Examination	20	20	10
	Internal Examination	**20**	**20**	**10**
	Total Internal Marks		**60**	**30**
Semester Examination			**40**	**20**

5. Recommended Books:

- Ian Sommerville (2016). Software Engineering, 10th Edition, Pearson.
- Roger S. Pressman and Bruce R. Maxim (2019). Software Engineering - a practitioner's approach, 9th Edition, McGraw-Hill Education.

UI/UX Design

Course Code	NC	**Year/Semester**	II/II
Credit Weightage	3	**Lecture**	3 hrs/wk
		Tutorial	1 hrs/wk
		Practical	2 hrs/wk
		Total	96 hrs

1. Course Description

This course will teach you the fundamental process of experience design as well as how to effectively evaluate your work with the people you are designing for. You'll learn fundamental design research methods that will help you understand people, the sequences of their actions, and the context in which they work. You will learn practical techniques for making sense of what you see and transforming your observations into meaningful actionable insights and unique opportunity areas for design through the assignments.

2. Course Objectives

The course readings, activities, and assignments were designed to give students opportunities to meet the following objectives by the end of the semester:

- To create and document a portfolio-quality concept design using well-known experience design methods.
- To select appropriate methods and theories based on an understanding of the strengths and limitations of available approaches.
- To articulate your approach to design and user experience design specifically in a clear manner.
- To describe popular research and design processes used in UX as well as alternative approaches to UX design.
- To implement design prototypes in tools such as Photoshop, Illustrator, Figma, Adobe XD, HTML, CSS, Sketch, and Axure.
- To experiment with peer design critique.

3. Course Details

3.1 Theory (48 hrs)

Unit 1: **Introduction to User Experience** **(6 hrs.)**

User Interface Design, User Experience Design, UI vs UX, User-centric Design, Key framework of UX design, Universal and inclusive design, UX design tools

Unit 2: **User Experience Design** **(10 hrs.)**

Users, empathy map, users story, edge cases, users journey map, storyboarding, psychology principles that influence design, Ideation: exploring ideas, business needs and competitive analysis.

Unit 3: **User Experience Research** **(12 hrs.)**

UX research, power of UX research, bias in UX research, choosing the right research method, research plan, goals and questions, UX research process, ethics, privacy and data in real world, usability studies, its types and process.

Unit 4: **Prototyping** **(12 hrs.)**
Prototypes: wireframes and low-fidelity prototypes, introduction to Figma, digital wireframes, Lo-Fi prototype, usability testing and iteration on design.

Unit 5: Visual Design/ UI Design **(8 hrs)**
Introduction to Visual Design/ UI Design, Design Principles: Contrast, Visual hierarchy, Alignment, White space, Gestalt principle, Colour theory, Typography, layouts, Iconography and symbols, transition from Lo-Fi to Hi-Fi Design, Design System.

3.2 List of Tutorials

- Navigating the Figma User Interface Designing a Web Application
- Working with UI Elements and Content Making a Design Responsive
- Reusing Design Elements
- Building Case Studies
- Refining User Research and Creating Personas
- Designing Login Pages using Figma
- Developing Professional Website

3.3 Laboratory Works

Mini Research Project
Your mini project for this course is an applied piece of user experience research.
This may be:

- An investigation of user needs for a proposed software product, information system, web site, or similar; or
- An evaluation and assessment for a prototype or an existing system or service.

The more real it is, the better the project. The best projects are for an actual client. But you can also pick a site or product with which you have no direct connection.
Given the constraints of the semester, this is likely to be more of a pilot project than an actual, completed project for a client. It is most important that you get to practice a variety of methods and the entire process from planning to analysis and delivery of a product. It is unlikely that you will have time to go into enough depth with each method to have defendable results.This can be an individual or a group project; group projects are recommended.

4. Evaluation Scheme

Evaluation of student's performances is divided into two main parts: (1) Internal Evaluation and (2) Semester examination, with breakdowns as follows:

Evaluation	**Components**	**Weightage**	**Full Marks**	**Pass Marks**
Internal Evaluation	Class Attendance and Performance	5	20	10
	Assignment	10		
	Seminar/Project/Presentation	5		
	Practical/Lab Examination	20	20	10
	Internal Examination	**20**	**20**	**10**
	Total Internal Marks		**60**	**30**
Semester Examination			**40**	**20**

5. Books

- Elizabeth Goodman, Mike Kuniavsky, 2012. Observing the User Experience: A Practitioner's Guide to User Research, 2nd edition.
- Designing the User Interface, Ben Schneiderman, McGraw Hill Edition, 4th Edition.
- I. Scott MacKenzie. 2013. Human-Computer Interaction: An Empirical Research Perspective. Morgan Kaufmann
- Don Norman. 2013. The Design of Everyday Things: Revised and Expanded Edition. Basic Books.
- Tomer Sharon, 2016. Validating Product Ideas.